Tyler James

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Education

University of Kansas — Lawrence, KS
Bachelor of Science in Computer Engineering — *GPA: 3.92 — Expected May 2027*

Experience

University of Kansas – Computer Systems Lab — August 2026 – Present
Undergraduate Research Assistant — *Lawrence, KS*

- Conducting research under Professor Heechul Yun on machine learning, computer vision, and autonomous flight.

Textron Aviation — May 2026 – August 2026
Product Engineering Intern — *Wichita, KS*

- Automated regression testing for Garmin avionics systems using Python, improving repeatability of validation across Cessna and Beechcraft aircraft platforms.
- Produced wiring diagrams and harness designs in Siemens Mentor Graphics to integrate avionics components and electrical interfaces into a new test bench.
- Modeled and simulated sensor-emulation circuits in LTspice and Altium Designer to reproduce aircraft sensor inputs for avionics test-bench validation.

CRB — May 2025 – August 2025
Control Systems Integration Intern — *St. Louis, MO*

- Programmed, tested, and debugged PLC logic in Rockwell Studio 5000 for a new Kikkoman manufacturing facility, including safety interlocks and process-control sequences.
- Developed production-ready HMI screens in FactoryTalk View for multiple process subsystems, emphasizing operator workflow and real-time system visibility.

Leadership

Theta Tau — Zeta Chapter — August 2025 – Present
Treasurer | Former Professional Development Chair — *Lawrence, KS*

- Managing a $7,000+ budget for a 70+ member professional engineering fraternity; previously increased professional development event attendance by 5%.

Projects

FPGA-Based Digital Waveform Generator | *Xilinx Vivado, VHDL, Oscilloscope* — April 2026 – May 2026

- Implemented a variable-frequency digital waveform generator in VHDL using a programmable accumulator to produce 8-bit digital waveform samples.
- Constructed a resistor-based DAC to convert digital samples to an analog output and validated generated waveforms using a digital oscilloscope.

Tenor: Embedded ML Audio Detection System | *Edge Impulse, ESP32, Raspberry Pi, Qt* — April 2026

- Built a low-cost embedded audio anomaly detection system using an ESP32, transmitting detected events over Wi-Fi to a Raspberry Pi-based monitoring hub.
- Trained and deployed an audio classification model using Edge Impulse and implemented a Qt-based interface for real-time visual alerts.

LED-Powered Digital Clock | *C++, ESP32, Fusion 360, 3D Printing* — August 2024 – February 2025

- Integrated an ESP32, custom seven-segment LED arrays, and a real-time clock module into a custom-designed and 3D-printed enclosure.
- Developed a C++ LED driver using FastLED and interfaced with the real-time clock over I2C to retrieve time data and update the display.

Technical Skills

Languages: Python, C/C++, VHDL, Ladder Logic
Software: Xilinx Vivado, Altium Designer, LTspice, Rockwell Studio 5000, FactoryTalk View, Siemens Mentor Graphics, Fusion 360, Qt Designer, Git, Linux, Docker, Proxmox
Hardware: Arduino, ESP32, FDM 3D Printing, Circuit Design, Oscilloscopes, FPGA Development